

Deploy resilient edge computing with AI-driven anomaly detection to enhance energy efficiency and security in critical infrastructure.

Energy × Energy optimisation × Edge computing · OS075 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
47 is the world moving	52 can we play, can we win	MEDIUM 0 named in the evidence	€94.1m partial evidence · per year	LATER research and standards activi...	L1 13 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 3.0 published reference(s) in Energy, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity is about deploying resilient edge computing with AI-driven anomaly detection to improve energy efficiency and security in critical infrastructure. It targets energy sector operators who need real-time, low-latency monitoring and response at the edge, driven by increasing cyber threats and environmental challenges. The time is now because edge AI technologies have matured to run on resource-constrained devices, and EU-funded research like the RESCUE project is validating the approach.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€114m €18.0m – €1.8bn	€94.1m €14.9m – €1.5bn	€1.1m €178k – €17.4m	partial
Observed: tendered contract value, annualised	€172m €172m – €460m	€172m €172m – €460m	€2.1m €2.1m – €5.5m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Energy (EU27_2020)	5,796 enterprises	observed	Eurostat · sbs_sc_oww	2024
Enterprises adopting Edge computing	28.4 % of enterprises (10+ employees)	proxy	Eurostat · isoc_cicce_usen2 · E_CC_PCPU	2025
Annual value of one engagement	€299k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Size-weighted buyer base	1,341 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Cloud computing services by NACE Rev. 2 activity series E_CC_PCPU is used as a proxy for Edge computing: Cloud compute purchase as nearest series. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

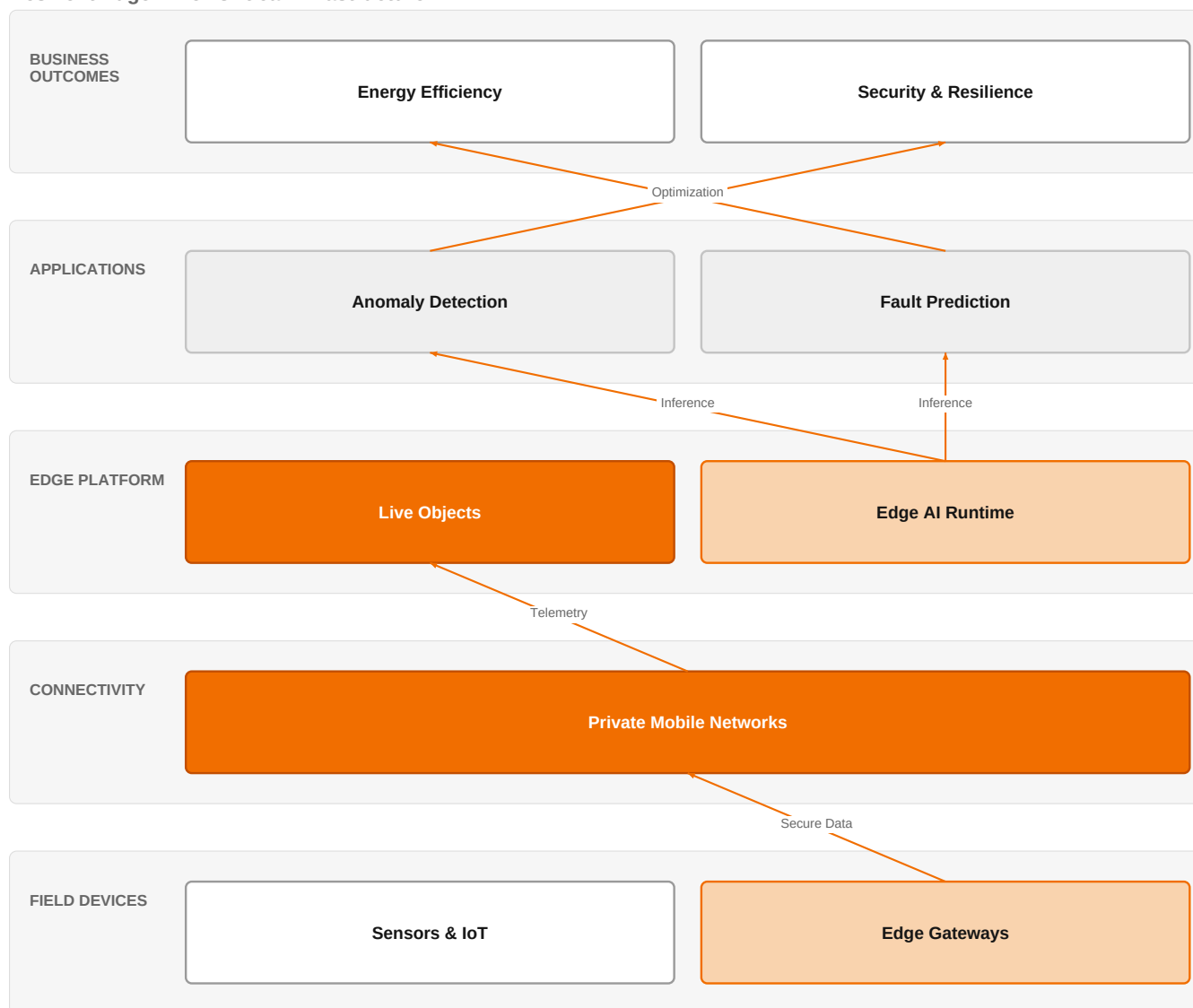
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€172m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 21 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

Resilient Edge AI for Critical Infrastructure



This diagram shows how Orange's connectivity and IoT platform, combined with partner edge AI, deliver real-time anomaly detection for energy and security outcomes.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Edge AI has matured to the point where neural networks can run on resource-constrained microcontrollers, enabling real-time, on-device processing with reduced cloud dependency (SIG-0860D0B1EBA0). However, deploying deep learning models on industrial edge devices still faces challenges in balancing inference accuracy and low latency (SIG-5B350B1A3B15). The increasing scale and heterogeneity of IoT systems introduce uncertainty that conventional deterministic optimization cannot handle, prompting robust co-optimization of power allocation and computation offloading (SIG-D8AEDBAD839). Traditional cloud-based solutions suffer from excessive latency and energy overhead, driving the shift to edge-based AI for early fault detection (SIG-F48074B55B20). The RESCUE project is specifically strengthening the resilience, security, and adaptability of critical infrastructure through advanced edge computing and AI-driven anomaly detection, in response to increasing cyber threats and environmental challenges (SIG-F292FB8C24B6).

Sources: SIG-0860D0B1EBA0 openalex.org 2025-12-05; SIG-5B350B1A3B15 Journal of Organizatio 2026-05-21; SIG-D8ADEDBAD839 Elsevier BV 2026-01-01; SIG-F48074B55B20 openalex.org 2025-10-16; SIG-F292FB8C24B6 cordis.europa.eu 2025-09-01

Who buys, and why now

The buyer is typically the head of operations or the CISO at an energy utility or critical infrastructure operator. The pain is operational: they need to detect anomalies (e.g., equipment faults, cyber intrusions) in real time to prevent downtime or security breaches, but cloud-based analytics introduce latency and energy overhead. The trigger is often a near-miss incident, a regulatory audit, or a directive to improve resilience and energy efficiency. They are also motivated by the need to reduce energy consumption of their monitoring systems themselves.

Why it is hot now — every claim bound to a dated source

- The RESCUE project is strengthening the resilience, security, and adaptability of critical infrastructure systems through advanced edge computing. [SIG-F292FB8C24B6]
- Edge computing with AI-driven anomaly detection is being developed to address increasing cyber threats and environmental challenges. [SIG-F292FB8C24B6]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its Live Objects IoT platform to manage device connectivity and data ingestion, and its Private mobile networks to provide secure, low-latency connectivity at the edge. Orange Group researchers would contribute expertise in edge AI and anomaly detection. The engagement would start with a pilot on a specific site, deploying edge nodes (possibly using partner hardware from Dell or HPE) running AI models for anomaly detection, integrated with Orange's cybersecurity experts to ensure security. The solution would be certified against ISO 27001 and SecNumCloud where relevant, and could leverage Orange's defense and security experts for critical infrastructure requirements.

Why Orange can win

Orange can win because it combines connectivity (Private mobile networks), IoT management (Live Objects), and security expertise (cybersecurity experts, ISO 27001, SecNumCloud) in a single offer. The ability to provide a sovereign, certified solution is particularly valuable for critical infrastructure. However, the edge computing hardware and AI model optimization are not core Orange assets, so Orange would need to rely on partners like NVIDIA, Dell, or HPE for those components, which could dilute the differentiation.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L1	offer addresses the use case but not this technology	unconfirmed
Private mobile networks	offer	L1	offer provides the technology in a matching domain	unconfirmed
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
Dell	partner	L2	partner provides the required technology	unconfirmed
HPE	partner	L2	partner provides the required technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
Cybersecurity experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 7.1; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells Edge computing; competes in OX: Smart Industries — also an Orange partner	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	sells Edge computing; competes in Cybersecurity, OX: Smart Industries	structural	—
Lumen Technologies	Telecom operator peer	sells Edge computing; competes in Cybersecurity	structural	—
NTT DATA	Global systems integrator	sells Edge computing; competes in OX: Smart Industries	structural	—
Schneider Electric	Industrial / OT specialist	sells Edge computing; active in Energy; competes in OX: Smart Industries	structural	—
HPE (Aruba, Athonet)	Network equipment vendor	sells Edge computing — also an Orange partner	structural	—
Nokia	Network equipment vendor	sells Edge computing; active in Energy; competes in OX: Smart Industries	structural	—
Capgemini	Global systems integrator	active in Energy; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What specific operational pain points are you trying to solve with edge computing—latency, energy efficiency, security, or all three?
- 2 Do you currently have any edge computing deployments? If so, what are the main challenges you face?
- 3 What is your current approach to anomaly detection in your critical infrastructure, and what are its limitations?
- 4 Are you subject to any regulatory requirements for data sovereignty or security certifications that would influence your choice of solution?
- 5 How do you currently manage the connectivity for your distributed sites, and what are the reliability requirements?
- 6 What is your timeline for implementing a solution, and do you have a specific site or use case in mind for a pilot?

Objections you will meet

They will say	You can answer
We already have a cloud-based analytics solution that works well enough.	That's true, and cloud analytics have their place. However, for real-time anomaly detection at the edge, cloud solutions introduce latency and energy overhead that can be critical. Edge AI can process data locally, reducing response times and bandwidth costs. Let's discuss your specific latency and reliability requirements.
Edge computing is too complex and expensive to deploy across our infrastructure.	Complexity is a valid concern, but with our Live Objects platform and private mobile networks, we can manage devices and connectivity centrally, reducing operational overhead. We can start with a pilot on a single site to demonstrate value before scaling. Our partners provide cost-effective hardware options.
We are concerned about security and data sovereignty.	Security is a top priority for us. We offer ISO 27001 and SecNumCloud certified solutions, and our cybersecurity experts can ensure your edge deployments are secure. We can also provide sovereign infrastructure options to meet regulatory requirements.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for an energy-sector tender that combines Live Objects and Private mobile networks with cybersecurity expertise, referencing ISO 27001 and SecNumCloud certifications, and outline how NVIDIA, Dell, or HPE partnerships could enhance the solution.
Sales	In a conversation with an energy utility's head of operations, mention that Orange is strengthening critical infrastructure resilience through edge computing and AI-driven anomaly detection, and ask if they are facing challenges with cyber threats or environmental disruptions.
Strategist / Innovator	Commission a deep-dive brief on how the RESCUE project's edge computing approach can be paired with Live Objects and Private mobile networks to create a differentiated energy-sector offer, and identify which Orange Group researchers to engage.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-5B350B1A3B15	2026-05-21	Journal of Organizational...	1	Latency Trade-Offs of a Lightweight YOLOv11 Detection Algorithm on Edge Computing Nodes for Fac...
SIG-3772E07403F3	2026-01-02	openalex.org	1	Cloudburst AI: Orchestrating SRE-Driven AI Inference Across Multi-Cloud and Edge Zones
SIG-D8AEDBAD839	2026-01-01	Elsevier BV	1	Robust Energy–Latency Co-Optimization for Power Allocation and Computation Offloading in Edge C...
SIG-AECDD651DA98	2025-12-17	openalex.org	1	Edge-AI integrated secure wireless IoT architecture for real time healthcare monitoring and fed...
SIG-94BF1B359033	2025-12-11	openalex.org	1	Edge AI in Practice: A Survey and Deployment Framework for Neural Networks on Embedded Systems
SIG-0860D0B1EBA0	2025-12-05	openalex.org	1	Deploying TinyML for energy-efficient object detection and communication in low-power edge AI s...
SIG-8EC0D797B8D2	2025-10-27	openalex.org	1	AI-Powered Security for IoT Ecosystems: A Hybrid Deep Learning Approach to Anomaly Detection
SIG-D4C81A60CDDC	2025-10-17	openalex.org	1	An Energy-Aware Generative AI Edge Inference Framework for Low-Power IoT Devices
SIG-F48074B55B20	2025-10-16	openalex.org	1	Implementation of edge AI for early fault detection in IoT networks: evaluation of performance ...
SIG-F292FB8C24B6	2025-09-01	cordis.europa.eu	1	Resilient Edge Systems for Critical infrastructure and Urban Environments
SIG-D4CA61FE83AD	2025-08-29	openalex.org	1	AI-powered cyber resilience: a reinforcement learning approach for automated threat hunting in ...
SIG-B7834050E076	2025-08-26	openalex.org	1	A Case Study of Environmental Footprints for Generative AI Inference: Cloud versus Edge
SIG-17DA3B724138	2026-04-27	arxiv.org	3	System-aware contextual digital twin for ICS anomaly diagnosis

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); risks_and_unknowns (missing from the model output); diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:56:43+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.